PHY101: Introduction to Physics-1

Monsoon, 2018

Tutorial-15

1. In which case will the efficiency of a Carnot cycle be higher: when the hot body temperature is increased by ΔT, or when the cold body temperature is decreased by the same magnitude?

2. A refrigerator is run by 1 KW electric motor of 60% efficiency; if the machine is working between 273 K and 293 K; find the time required to freeze 9 kilogram of water. Given the latent heat of ice = 80 Cal/gm and 1 cal $=4.2\times10^{7}$ erg.

3. 4. Calculate the work done by one mole of a gas during the isothermal expansion from an initial volume V_i to final volume V_f when the equation of state is,
(a) $P(V - b) = RT$ where R & T are constants
(b) $PV = RT\left(1 - \frac{B}{V}\right)$ where B is the function of temperature.

(4) (a) Calculate the maximum work in Joule obtainable from a heat reservoir consisting of 200 Kg of iron heated initially to a temperature of 1500°C, using the ocean at 12°C as the second heat reservoir. Assume that the specific heat capacity of iron is constant and equal to 0.60 Joule/gram-deg. (b) Calculate the entropy change of the universe in this process.